

Element H: Prototype Testing and Data Collection Plan

To start, it's crucial to test the safety and durability of our prototype, especially since it will be used by children ages 4-6. We want to ensure that the product is safe and won't cause any harm. To assess its safety, we'll conduct tests outside of a child's hands to minimize risk to them. Key safety tests will include:

Safety 1: Edges Test

We will have groups of teachers run their fingers along the edges of the prototype to check for any scratches or cuts. Although we've smoothed the edges, this precautionary test will help us confirm that the design is safe for little hands.

Safety 2: The 'Throw' Test

Understanding that kids can be rough with their toys, we need to ensure our product can withstand some impact. We'll have three individuals of varying strength slam the prototype onto the floor to see if it can endure the force without breaking.

Durability 1: Stress Test

To ensure our product lasts, we'll perform a series of stress tests. We'll stack different weights—5, 10, and 15 pounds—on top of the prototype to see if it can hold up under pressure without collapsing.

Easy to Comprehend 1: Instructions Test

To evaluate how well kids can understand and use the Makey Makey kits, we'll conduct tests with different control groups, all consisting of children ages 4-6 who have no prior experience with the kits. While we know that every child learns differently, this approach will help us gauge the effectiveness of various instruction methods. Control Group A will receive only written instructions on how to set up and use the Makey Makey kit. Control Group B will get written instructions along with a live demonstration from a teacher, showing them how to connect the kit to different objects and create sounds. Control Group C will have written instructions plus a fun, interactive video that explains how to use the Makey Makey playfully. We'll document the results after each child tries the kit three times to see which method helps them understand and engage with the Makey Makey best.

Note: Due to the area we are from, there are not many experts to contact in regards to our tests, so I will add that we contacted a children's toy manufacturer, however, they never reached out to us for a follow up.

Expert #1: Mrs. Staci Griffey, Teacher

- Her feedback was all positive, and she said, "They are incredibly engaging and do a great job of sparking creativity in kids, making learning about electronics fun. However, she noticed that some instructions could be clearer, especially for younger users. Simplifying the language or adding more visuals could help. Overall, she believes they're a fantastic tool for encouraging hands-on learning!"

Expert #2: Mrs. Michelle Baker, Culinary Teacher

- Her feedback was about the same as Mrs. Griffey's, she also believes they're engaging and do a great job at sparking interest and creativity in kids, and making learning about electronics fun. However, she also noticed that some instructions could be easier, especially for younger kids.